



Shri Vile Parle Kelavani Mandal's
Shri Bhagubhai Mafatlal Polytechnic
And College of Engineering



PRESENTS

HACKATHON PROBLEM STATEMENTS



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DOMAIN 1

FULL STACK DEVELOPMENT (FSD)



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FSD 1

DIGITAL IDENTITY SYSTEM FOR GIG WORKERS

The rapid growth of the gig economy has created diverse employment opportunities across delivery, transportation, home services, freelancing, and other digital platforms, but workers often operate across multiple platforms where their earnings, work history, skills, ratings, and professional reputation remain fragmented. This makes it difficult for workers to present a complete and verifiable professional profile and can limit their access to financial services, employment opportunities, and other benefits. Develop a secure and portable full-stack Digital Identity and Reputation Platform for Gig Workers that consolidates verified information from multiple work platforms into a unified digital profile, enabling workers to securely demonstrate their skills, experience, earnings history, and professional reputation while giving authorized organizations a reliable basis for evaluating their professional and financial profile.



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FSD 2

SMART CIVIC ISSUE REPORTING & RESOLUTION PLATFORM

Citizens frequently encounter civic issues such as potholes, damaged streetlights, overflowing garbage, water leakage, and road damage, while authorities often struggle with duplicate complaints, manual prioritization, fake reports, and inadequate verification of resolutions. Develop a full-stack Civic Issue Management Platform that allows citizens to report problems using photographs and geolocation, automatically clusters duplicate reports using location and image similarity, and calculates an intelligent priority score based on factors such as duplicate-report count, proximity to schools or hospitals, and complaint age. The platform should enable authorities to verify, assign, track, and resolve complaints, requiring after-repair photographs and using image comparison and/or citizen confirmation to verify the resolution, while detecting suspicious or spam reporting patterns and providing real-time status updates and dashboards to monitor complaints, priorities, assignments, and resolution progress.



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FSD 3

SMART APPOINTMENT QUEUE MANAGEMENT SYSTEM

Customers and citizens often spend significant time waiting in queues at clinics, service centers, educational institutions, banks, and other facilities, while organizations may struggle to manage appointments and walk-in demand efficiently. Develop a full-stack appointment and virtual queue management platform that allows users to book appointments or join digital queues, receive estimated waiting times and notifications, while service providers can manage counters, appointments, queue priorities, and real-time service status through an administrative dashboard.



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FSD 4

CODE MAFIA: MULTIPLAYER COLLABORATIVE DEBUGGING CHALLENGE

Software development relies on collaboration, debugging, and teamwork, but identifying whether a team member is unintentionally introducing problems or deliberately causing them can be challenging. Develop a full-stack multiplayer coding and debugging game in which players collaboratively work on a shared, intentionally flawed software project while some players are secretly assigned the role of Mafia. Developers must identify and fix bugs, stabilize the application, and pass defined tests, while Mafia players attempt to disrupt progress by introducing bugs, creating regressions, or preserving existing issues without being detected. The platform should support game creation and configuration, hidden role assignment, real-time collaborative coding, controlled code execution and testing, player activity monitoring, voting and elimination, synchronized game-state management, and configurable victory conditions, providing an engaging environment that combines software development skills, debugging ability, collaboration, observation, and strategic decision-making.



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DOMAIN 2

ARTIFICIAL INTELLIGENCE(AIM)



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AIM 1

AI-DRIVEN PERSONALIZED CAREER COUNSELLING

Many students choose career paths based on societal expectations, peer influence, parental pressure, or limited awareness of available opportunities rather than their individual aptitude, interests, personality, skills, and preferences, often resulting in career dissatisfaction, low motivation, poor performance, and frequent career changes. Develop an AI-powered personalized career counselling platform that evaluates a student's strengths, interests, personality, skills, and preferences to recommend suitable career pathways while considering current industry trends, job-market demand, required competencies, educational programs, and learning pathways, enabling students to make informed and personalized career decisions.



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AIM 2

AI-POWERED FINANCIAL AUDIT ANOMALY DETECTION PLATFORM

Chartered Accountants, particularly those managing multiple SME audits, often spend considerable time manually reviewing ledgers, trial balances, and transaction records to identify potential financial irregularities before conducting detailed audit procedures. This process can make it difficult to efficiently detect patterns such as duplicate transactions, round-tripping, backdated entries, GST-to-book mismatches, unusual month-end spikes, and other anomalous activities. Develop an AI-powered audit analytics platform that automatically analyzes financial data, detects and prioritizes suspicious transactions and financial patterns, and provides explainable, risk-based insights to help Chartered Accountants focus their audit efforts on high-risk transactions, accounts, and areas requiring further investigation.



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AIM 3

AI-POWERED MISINFORMATION DETECTION AND CREDIBILITY ASSESSMENT SYSTEM

The rapid growth of digital media and social networking platforms has made it increasingly difficult to distinguish authentic information from fake, misleading, or manipulated content. The high speed and volume of online information can enable misinformation to spread widely, potentially influencing public opinion, creating social unrest, and reducing trust in reliable information sources, while traditional manual fact-checking methods are often too slow to handle this scale.

Develop an AI-powered misinformation detection and content verification system that analyzes news articles, social media posts, and online claims to assess content credibility and classify it as genuine, misleading, fake, or potentially manipulated, while providing users with understandable evidence or indicators supporting the assessment.



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DOMAIN 3

SOCIAL IMPACT CHALLENGES (SIC)



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SIC 1

CRISISCONNECT – REAL-TIME EMERGENCY ASSISTANCE RESPONSE PLATFORM

Develop a centralized emergency assistance and community response platform that connects people in urgent need with volunteers, NGOs, and authorized organizations during natural disasters, accidents, medical emergencies, and other crisis situations. The platform should enable users to create, verify, discover, and track requests for essential assistance such as blood, food, medicines, oxygen, shelter, transportation, and rescue services, while providing real-time status updates, location-based matching, notifications, and verification mechanisms to prevent duplicate or outdated requests and ensure that available resources and volunteers are directed toward genuine and active emergencies.



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SIC 2

MANDIMITRA – SMART CROP PRICE SELLING DECISION SUPPORT SYSTEM

Develop a farmer-friendly crop price decision-support system that helps farmers make informed decisions about when and where to sell their produce by providing reliable and timely mandi price information. The platform should allow farmers to select their crop and nearby market, retrieve current and historical market prices from reliable sources such as Agmarknet, analyze recent price trends, and present a simple actionable recommendation such as “Sell Today” or “Wait 2–3 Days,” helping farmers reduce dependence on middlemen, improve price awareness, and potentially increase their returns.



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SIC 3

CAREERMITRA – RURAL STUDENT CAREER OPPORTUNITY NAVIGATOR

Many rural students make career decisions based on limited local exposure and may be unaware of the wide range of career pathways, scholarships, entrance examinations, courses, internships, and higher-education opportunities available to them. Develop a simple, multilingual career guidance platform that understands a student's interests, academic background, skills, and aspirations and helps them discover suitable career pathways and opportunities. The system should provide personalized recommendations, eligibility information, learning roadmaps, scholarship discovery, and actionable guidance to help students make informed educational and career decisions.